



JAIN COLLEGE OF ENGINEERING AND RESEARCH BELAGAVI

Department Of Computer Science And Engineering

Project on

“Learn-sphere”

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Introduction

Learn Sphere is an online learning platform designed to connect trainers and students in a simple and effective way. It allows trainers to create and manage courses, while students can explore, purchase, and engage with educational content. The platform is built to be user-friendly and interactive, making learning easy and accessible. Powered by technologies like Spring Boot, MySQL, and Razorpay, it ensures a secure and smooth experience for all users. Learn Sphere promotes knowledge sharing and engagement, making it a valuable tool for both teaching and learning.

Application

- ❖ **Create a user-friendly learning platform:** A user-friendly learning platform offers easy access to courses and progress tracking for students and instructors.
- ❖ **Allow students to enroll and access content:** Students can easily enroll in courses and access course content anytime.
- ❖ **Allow trainers to manage and post courses:** Trainers can create and organize courses, adding lessons, materials, and assignments. They can also update or manage course content as needed.
- ❖ **Support secure user registration and login:** The platform ensures secure user registration and login through encryption and authentication protocols.

Advantages and Disadvantages

Advantages

- **Cost-Effective:** Many online courses are more affordable than traditional education, and there are often free resources available.
- **Accessibility:** Learners from anywhere can access courses, breaking down geographical and time barriers.
- **Wide Range of Courses:** Online platforms offer a variety of courses from different fields, providing more learning opportunities.

Disadvantages

- **Lack of Social Interaction:** Online learning can feel isolating without face-to-face interaction with peers or instructors.
- **Technical Issues:** Connectivity problems, software glitches, or lack of access to necessary technology can hinder learning.
- **Potential for Distractions:** Learning from home or remote environments can introduce distractions, affecting focus and productivity.

System Architecture Overview

1. Frontend (Presentation Layer):

- ❑ The frontend uses HTML, CSS, JavaScript, and Thymeleaf to create a responsive, user-friendly interface.
- ❑ It communicates with the backend through REST APIs for smooth navigation by trainers, students, and admins.

2. Backend (Business Logic Layer):

- ❑ The backend, built with Spring Boot, manages core functions like authentication, course management, and payment processing.
- ❑ It provides RESTful APIs for the frontend to handle actions such as viewing, enrolling, and payments.

3. Database (Data Layer):

- ❑ MySQL stores data related to users, courses, and payments in tables for easy access.
- ❑ Spring Data JPA simplifies database interaction through a repository layer.

Result

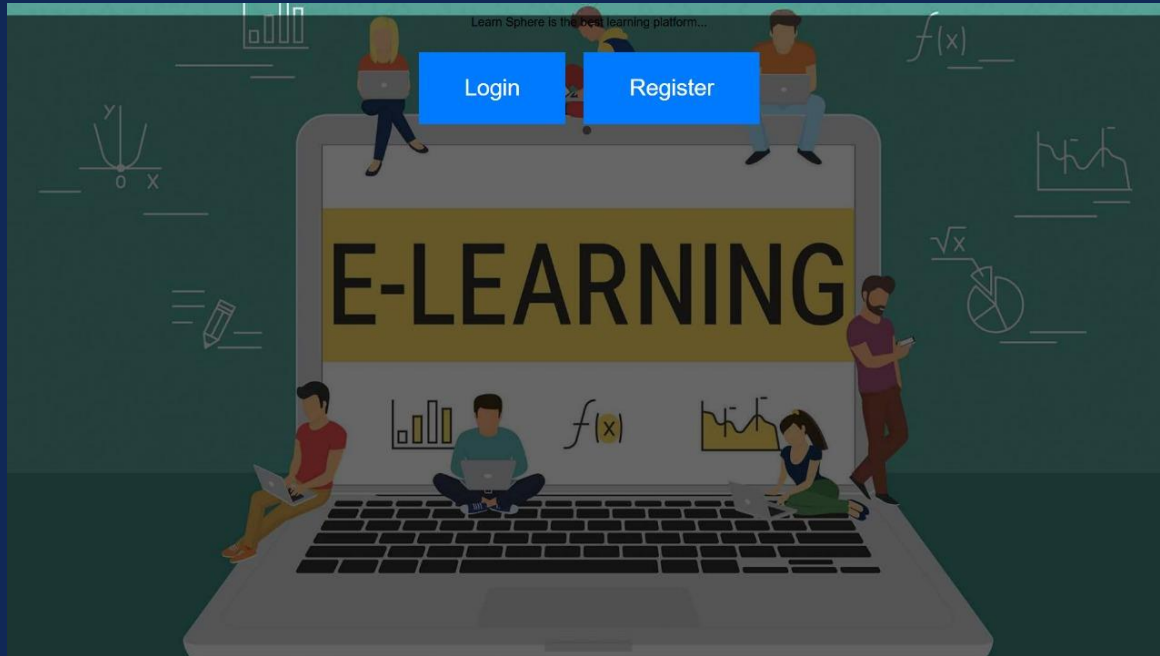


Fig.1: Home page

A screenshot of the "Learn Sphere" registration form. The form is titled "Learn Sphere" and contains the following fields: "Name" with the value "Jyothana kudachi", "Email address" with the value "jyo@gmail.com", and "Password" with masked characters "*****". Below the password field, there is a "Role:" section with two radio buttons: "Trainer" (selected) and "Student". A "Submit" button is located at the bottom right of the form. A small text line below the email field states: "We'll never share your email with anyone else."

Fig.2: Register page

Welcome trainer!

Add Course

Add Lesson

View Courses

Welcome Student!

Purchase Course

My Courses

Add a new lesson!

Course ID

02

Lesson ID

002

Lesson Name

Database of Schema Design

Lesson Topics

Fundamentals of Schema Design and Management

Lesson Video Link

https://youtu.be/hQvCOBv_-LE?si=SnhH-ikka0dsIGts

Add Lesson

Course Id	Course Name	Course Price	Lessons	Buy Course
1	Java	25000	Java Methods	BUY
2	SQL	10000	Database of Schema Design	BUY

Fig.3: Trainer access pages

Fig.4: student access pages

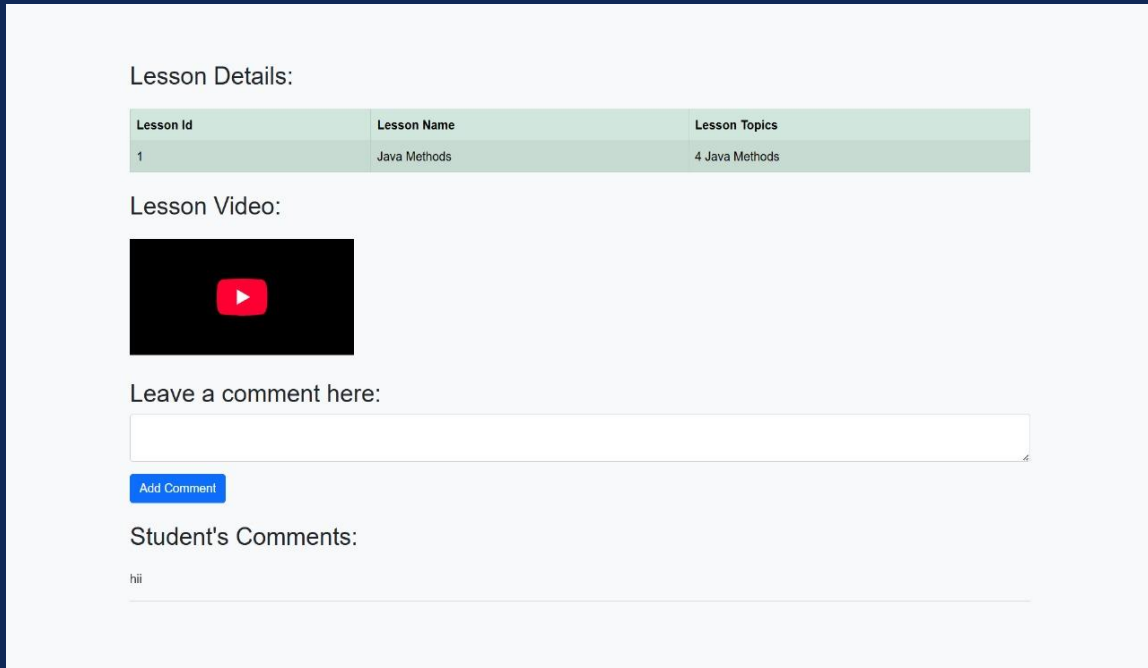


Fig.5: Lesson detail page

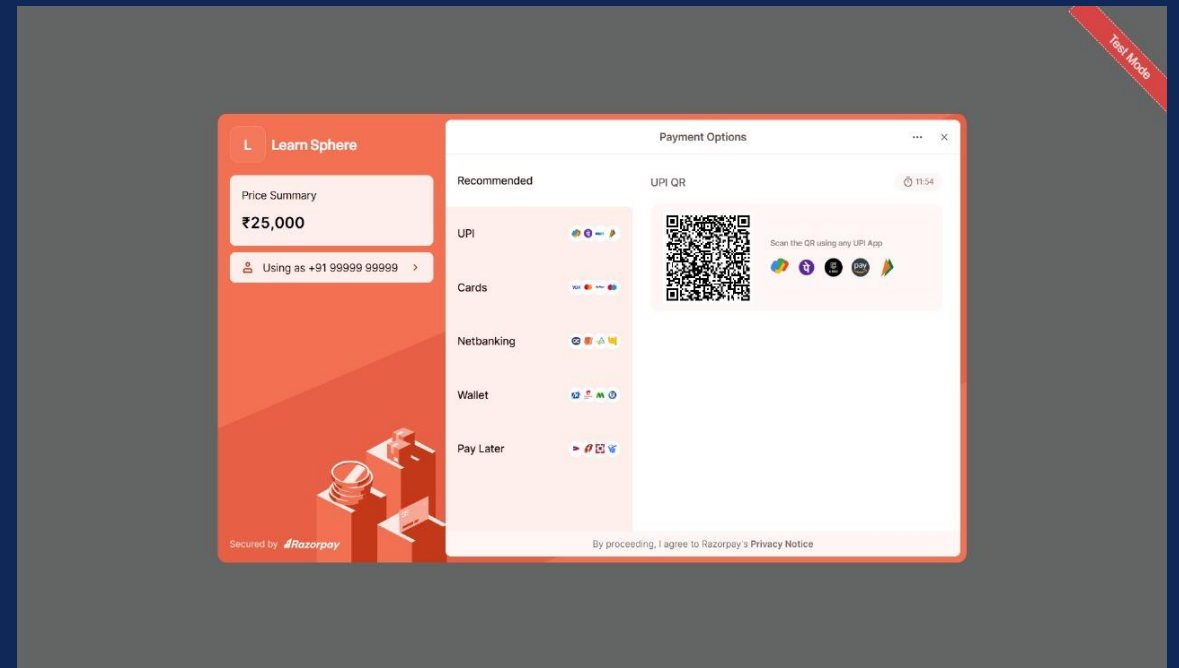


Fig.6: Payment access page

CONCLUSION

In conclusion, online learning platforms offer flexible, accessible, and cost-effective education for learners worldwide. While they present some challenges like reduced interaction and the need for self-discipline, their benefits make them a powerful tool for modern education and lifelong learning.

The background features a dark blue gradient on the right side, transitioning into a lighter blue and purple gradient on the left. Overlapping the lighter area are several large, soft-edged, organic shapes in various shades of purple and blue, creating a layered, abstract effect.

THANK YOU